Poetic Form Classification Using a BERT + BiLSTM Model

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Abstract

Poetic form classification presents an interesting challenge within the study of natural language processing due to the subjective and dynamic nature of poetry and poetic forms. In this work, we propose a hybrid architecture combining BERT with a BiLSTM layer to capture both contextual (i.e., themes and subject matter) and structural (i.e., stanza length or repetition and stress patterns) features of poetry. We trained and evaluated the model on a dataset of 2,270 poems across nine poetic forms, demonstrating that our hybrid approach achieves an F1-weighted-average score of 0.86 (F1-macro: 0.83). Analysis of our results reveals that unfixed forms, such as odes and ballads, which rely on imagery and themes to shape form, were challenging for the model to distinguish, whereas the patterns of fixed forms, such as sonnets and villanelles, were more easily learned.

1 Introduction

Poetic form pertains to the specific structure and rules that govern the construction of a poem. Such a definition implies strictness. But in reality, poetic forms are dynamic and sometimes not so rigidly defined, often sharing characteristics and blending together. While they emerge from specific historical and linguistic backgrounds, forms also move between cultures and adapt (Walsh et al., 2024). Key elements defining poetic forms include meter, rhyme scheme, the number of lines, and stanza structure. Form often enhances a poem's meaning—such as the pantoum's ability to evoke a circular feeling as themes and images return and evolve.

The intricate and overlapping nature of poetic forms along with the nuances of poetic language present an interesting challenge in the realm of natural language processing. A poetic form classification model must be able to capture the key elements that define one form and account for the subject matter that defines another. Even for fixed forms that follow established metrical and structural patterns, the artistic nature of poetry means these patterns can include subtle deviations that challenge computational analysis. Flexibility of form also makes it challenging to create definitive classifications; what one expert might call a sonnet, another might view differently (Walsh et al., 2024).

The automatic classification of poetic forms could prove useful in practical applications in library science and literary organizations. In library science, accurate form classification can streamline the discovery and curation of poetic works within large collections. For literary scholars, automated poetry analysis tools can facilitate large-scale studies of poetic evolution and form variation across different periods and traditions (Pozo, et al., 2023). An automated poem classification system also holds practical significance for professionals managing literary submissions, such as editors of poetry magazines (Ceccon, 2023).

This study explores how a pre-trained language model such as BERT combined with a secondary model handles classifying poetic texts based on the poem's designated form, focusing on nine poetic forms. The approach of using a BERT model for language understanding in tandem with a secondary BiLSTM model to capture structured patterns is motivated by the observation that form classification requires understanding both the linguistic content and the structural features of poetry.

2 Background

2.1 Previous Work

Research in the realm of poetry and natural language processing has a long, albeit not broad,

history. Earlier work focused primarily on rule-based systems for analyzing poetic features like rhyme and meter. Since the early 2010s, there has been significant progress from basic metrical analysis to more advanced modern machine learning (De Sisto et al., 2024). Specific to the task of classifying poems through machine learning methods, prior work centered on classifying the poems by sentiment or emotion. Another study focused on classifying poems by topic using a fine-tuned BERT model. The study tackled the issue of significant class imbalance within poetry datasets, demonstrating performance improvements through undersampling majority classes and implementing weighted loss functions (Ceccon, 2023).

Other recent research has shifted to examining how well LLMs can comprehend poetry. "Sonnet or Not" (Walsh et al., 2024) introduced the task of poetic form detection within the poetry/creative text generation and analysis space of NLP studies. The study results revealed that LLMs demonstrated competence in identifying fixed poetic forms, such as sonnets, but showed particular weakness in analyzing unfixed forms, i.e., those dependent on visual or thematic elements (Walsh et al., 2024). In this work, we build off their introduction of the task of identifying poetic form and focus on the development of a hybrid model to classify poetic forms rather than on evaluating LLM's ability to understand or detect poetic form.

Further work on metrical analysis was completed by the author of "Metrical Tagging in the Wild." Haider approached metrical analysis as a sequence classification task, where syllables served as input to predict metrical labels, showing that BiLSTM-CRF models utilizing syllable embeddings demonstrate superior performance compared to a CRF baseline and BERT-based methods. Based on Haider's use of syllable embeddings as part of the model to perform metrical tagging, we also incorporate syllable embeddings into the BiLSTM layer of our hybrid model to gain metrical information.

Other recent progress includes the development of domain-specific models with cross-linguistic capability. Noting the scarcity of tools to automatically analyze and scan poems, researchers developed ALBERTi, the first large language model specifically pre-trained for poetry analysis across multiple languages. Built upon multilingual BERT and trained on over 12 million verses across 12 languages, ALBERTi demonstrated sound performance in tasks like stanza classification and metrical pattern prediction.

A common thread through most of these works is the scarcity of comprehensive datasets and tools specific to poetry within the sphere of natural language processing. However, there are some researchers hoping to address that scarcity by developing new methods and technologies for improving poetry research. Our contribution to this field is the development of a hybrid BERT-BiLSTM model for poetic form classification.

3 Methods

3.1 Data

As highlighted by Ceccon, there is a significant gap in the availability of well-annotated and balanced datasets for NLP research in poetry (Ceccon, 2023). Datasets specific to poetic form seem to be even fewer; only two publicly available datasets of poems with labeled forms were identified for this study. Along with the introduction of the poetic form detection task, Walsh, et al., released a dataset of 1,453 annotated public-domain poems collected from the Poetry Foundation and Academy of American Poets websites. A Kaggle dataset of 6,300 poems and their form collected from Poetry Hunter was also found. In addition, poems labeled with a verse or stanza form were scraped from the Poetry Foundation website (2,387 poems). All poems (minus any duplicates) were combined into a data frame including the following metadata: author, title, poem text, form, topics (if provided), poem URL (if provided), and the website the poem was collected from.

Poems retrieved from the Kaggle dataset required cleanup since some of the poems' texts either had author notes at the end (which included dates or where the poem was written) or were written in a language other than English. Poems with text in a language other than English were removed; however, translations from another language to English were kept. If a poem had a few additional lines of text that were not part of the poem, those lines were removed by searching for

keywords, e.g., "Published online," and deleting such lines from the poem text string.

The combined dataset contains 141 unique forms. However, to limit the scope of this project and define the targets for this classification task, we focus on the poetic forms listed in *The Making of a Poem: A Norton Anthology of Poetic Forms*: villanelle, sestina, pantoum, sonnet, ballad, blank verse, elegy, pastoral, ode. The anthology also lists the heroic couplet and the stanza as forms, which have been excluded from this classification task since the combined and deduplicated dataset only has 11 labeled instances of a heroic couplet and only 13 labeled instances of a stanza. Irregular or alternate versions of each form were also excluded; for example, poems labeled as sonnets were included, but those labeled as Italian sonnets were not included. After filtering for the target forms, the dataset contained 2,270 poems.

There was a significant imbalance between sonnets and all other poems in the filtered dataset.

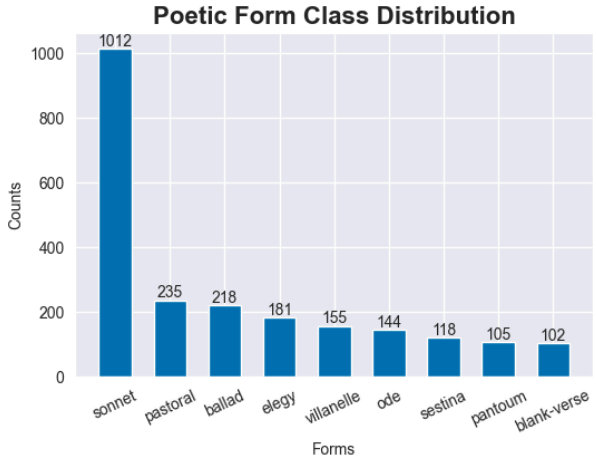

Figure 1. Poetic Form Class Distribution

To address this imbalance, undersampling was considered. In his study on imbalance, Ceccon had greater success with undersampling over using a weight factor for each class, but the advantage was slight. Since each poetry example holds distinct value, discarding so many sonnets would mean losing a wealth of information, as each poem could potentially reveal original linguistic patterns. With a limited total of poem examples, using class weights was deemed a better choice for this classification task since the strategy keeps all examples while still addressing the imbalance.

3.2. Classification and evaluation approach

This classification task focuses solely on the poems' textual content, excluding any information about author or other such metadata, as the source for predictions. Using topics as another data point to aid in the classification task was considered since some forms are based on subject matter, not meter, rhyme, or repetition. However, for many poems, the topic was unknown, or multiple topics were listed but lacked annotation of the primary topic. Because of these limitations, topic data was also excluded from the classification task.

For evaluation, we do not solely use accuracy metrics since the dataset is imbalanced. Evaluation metrics for this study also incorporate precision, recall, and F1 scores. Excellent performance is characterized by precision exceeding 0.80, recall surpassing 0.75, and F1 scores above 0.77, which seems ambitious for a poetic form classification task since forms can have aspects that might be hard for the model to detect. The aim of this work is to show that the hybrid model achieves F1 scores greater than 0.70 while maintaining balanced precision and recall scores along with consistent performance across poetic forms.

3.3 Models

Baseline model. The baseline for this study is a simple BERT model with no secondary model. The baseline builds off of the 'bert-base-cased' pre-trained model, taking the pooler output and adding one hidden dense layer with ReLU activation. After dropout regularization, a final softmax layer outputs class probabilities.

Model 1 – BERT + BiLSTM (no syllables). The next model for this study implements a hybrid BERT-BiLSTM model. The architecture again uses 'bert-base-cased'; however, instead of taking the pooler output, the sequence output of the BERT model is fed to the BiLSTM layer. Output from the BiLSTM layer is then passed through a dropout layer and then the classification layer.

Model 2 – BERT + BiLSTM + syllable features. This model uses the same foundational architecture as Model 1 but incorporates syllable features. The CMU Pronouncing Dictionary is used to extract syllable counts for each word in the input text. The sequence output from the BERT pre-trained model is concatenated with the syllable embeddings and

then processed through the BiLSTM layer. The model maintains the same subsequent layers as Model 1.

Model 3 – ALBERTi + BiLSTM + syllable features. Model 3 has the same architecture as Model 2 but builds off the ALBERTi pre-trained model rather than 'bert-base-cased'.

All models are compiled with Adam as the optimizer, categorical cross-entropy loss, and metrics for accuracy, precision, and recall. The models are initially built with the following hyperparameters: hidden layer size (default 256), dropout rate (default 0.3), and learning rate (default 0.00005). The max length used for all models is 512, the token limit in BERT.

4 Results and discussion

Baseline results. The baseline of a simple BERT model performed very poorly when evaluated on test data, producing an extremely low weighted average F1 score of 0.16 (macro average F1 score: 0.1). The low F1 score is a result of the model's low recall of 0.17 (macro average recall: 0.2). The model has oddly high weighted average precision of 0.41 (macro average precision: 0.14), indicating a conservative prediction pattern, that is, the model is only making predictions when it is very confident. Looking at the classification report heatmap reveals that the model was very conservative when predicting sonnets.

Figure 2. Classification Report Heatmap for Baseline Model

Though sonnets were the most frequent class in the dataset, the high precision score of 0.82 and low recall of 0.15 indicate that the model has learned to be fairly conservative with this class. Considering that class weights are being used to balance out this class's higher frequency, the weighting might be too aggressive, resulting in the model not predicting a sonnet when it is less certain of the form. The heatmap also reveals high recall but low precision scores for blank verse and sestinas along with pastorals. If we look deeper at the confusion matrix (Figure 3) for the model predictions, we see that the model is being too generous with these classes while failing to provide predictions for ballads, odes, and pantoums.

Such generosity is not overly surprising. The confusion matrix shows that villanelles and pantoums are frequently predicted as sestinas.

Figure 3. Baseline Model Confusion Matrix

These are forms that rely heavily on repetition. The key distinction between the forms is in their repetition patterns: sestinas repeat end words, villanelles repeat entire lines in a fixed pattern, and pantoums interweave lines between stanzas.

Blank verse appears to be the model's go-to prediction for elegies and odes. The model seems to be having trouble predicting forms that lack a fixed structure and rely more on subject matter to supply form. This behavior is the opposite of what was expected for a BERT model given its ability to capture context-aware meaning; however, distinguishing the nuances of and differences between ballads, elegies, odes, and pastorals may require a more complex model than our baseline provides.

Improvements to models. Though the weighting seems to have caused the baseline model to be conservative when predicting sonnets, the class weights were kept as is and combined with architectural improvements in Models 1 and 2 to maintain class balance while improving the model's ability to recognize more instances of the majority class and to help with the precision-recall imbalance of the blank verse and sestina classes. The primary improvement for both models was to add a BiLSTM layer to help the model pick up sequential patterns. Model 2 builds on Model 1, adding syllable embeddings as another input into the model to extract metrical features from the poem text. A limitation of the dataset was that the poems were not annotated with their meter, the basic rhythm and stress structure of a verse. In order to get syllable embeddings, the CMU Pronouncing Dictionary was used to obtain phonetic representations of words in an attempt to help the model infer metrical features.

Performance comparison. The performance of the baseline, Model 1, and Model 2 on test data were compared after training each model for one epoch using the same hyperparameters. Differences in how each model interacted with poem types and in metrics were examined to determine if the addition of syllable features impacted model performance.

Model 1 showed little improvement over the baseline, achieving a 32-percentage-point increase with a weighted average F1 score of 0.48 (macro average F1: 0.33). Model 2 was much better able to distinguish between pantoums, villanelles, and sestinas. The model was also less conservative in its predictions for sonnets, with the F1 score improving from 0.26 to 0.81. However, the same confusion between odes, elegies, and blank verse was observed.

Model 2 produced improved results compared to both previous models, achieving a weighted average F1 score of 0.66 and a macro average F1 score of 0.48.

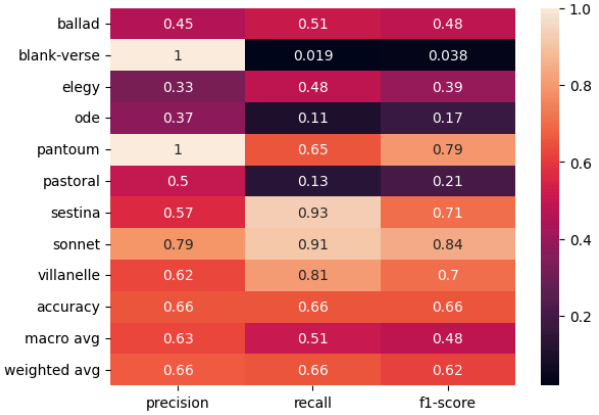

Figure 4. Classification Report Heatmap for Model 2 (1 Epoch)

The identical weighted average for precision and recall suggests that the model has achieved great balance between being too aggressive and too conservative in its predictions. However, based on the above heatmap, the model has learned to be extremely conservative with predictions for blank verse and may not be able to pick up on the form's distinct features. Also, the confusion matrix reveals that the model is primarily predicting two classes, sonnets (primarily being predicted for subject-matter-based forms and for blank verse) and odes (primarily being predicted for odes and blank verse). Confusion between sonnets and blank verse is telling since both are written in iambic pentameter; possibly, the model is picking up on the meter of these examples but has yet learned how to distinguish between the two forms. Overall, the metrics suggest the model is learning but hasn't converged given that the model has only seen the training data once.

Tuning Model 2. To further improve Model 2, the number of epochs was initially extended to three. The learning rate was also tested to make sure it was appropriate; a higher learning rate pushed the model to heavily favor one class and a lower

learning rate had very similar results to the original learning rate but produced more balanced precision and recall scores. On the next iteration of the model, the learning rate of 0.00005 was reduced to 0.000025 and the number of epochs was extended to five.

This final version of Model 2 shows marked improvement in performance over baseline. The extended training time allowed the model to learn subtle patterns to distinguish between forms. When evaluated against test data, the model achieved a much better balance between weighted average precision (0.87) and weighted average recall (0.86), resulting in an F1-weighted-average score of 0.86. Macro scores for precision, recall, and F1 all equaled 0.83. Drilling down into each class, predictions for villanelles (precision: 0.80, recall: 0.94), pantoums (precision: 1.0, recall: 0.93), and sonnets (precision: 0.90, recall: 0.92) show the most improvement. However, there are instances where the model is still struggling. Elegies appear to be troublesome for the model as it has high recall but lower precision, indicating that the model is overeager in predictions for this class, and perhaps features distinguishing this class from others might not be clear enough. Also, for pastorals, the model seems to be conservative when making predictions for this form.

Figure 5. Classification Report Heatmap for Model 2 (5 Epochs)

Model 2 and Model 3 comparison. For a third model to compare with Model 2, the pre-trained BERT model based on 'bert-base-cased' was swapped out with a BERT model using ALBERTi as a foundation. Since the ALBERTi model is trained specifically for poetry analysis, we believed Model 3 could potentially gain a more comprehensive understanding of poetic expression and perhaps better classify forms.

The ALBERTi model was initially trained for three epochs with a batch size of eight and a learning rate of 0.000025. When evaluated against the test data, the precision and recall scores of this initial model were fairly balanced for both the macro and weighted averages. However, there were certain forms that the model had trouble distinguishing; for example, the model often confused villanelles for pantoums and was unable to predict elegies.

Further training of Model 3 with the same hyperparameters as the final iteration of Model 2 produced similar results to Model 2's final results.

Figure 6. Classification Report Heatmap for Model 3 (5 Epochs)

Rather than struggling to distinguish elegies, Model 3 struggled with odes but showed some improvement in its predictions for pastorals over Model 2. Both models had stronger performance in forms of fixed structure but found it harder to predict forms defined by imagery or theme.

5 Conclusion

In this work, we developed a BERT + BiLSTM model for poetry form classification with weighted average and macro F1 scores greater than 0.80, along with balanced precision and recall scores. F1 scores by class were fairly consistent within the range of 0.70 to 1.0, but precision and recall scores for certain forms dipped below 0.70. Similar to the results in "Sonnet or Not", the model was successful in classifying poems of fixed form but struggled with poetic forms defined by subject matter. We should note as well that our work was

limited to only nine poetic forms and does not take into account other poetic forms that may require different techniques to obtain successful results. We also did not study the adequacy of the ALBERTi model without the combination of a BiLSTM and leave that for future study. Future research could also broaden the scope of this classification task to include other poetic forms and irregular versions of the forms we chose for this task.

Datasets

1. https://github.com/maria-antoniak/poetry-eval
2. https://www.kaggle.com/datasets/michaelarman/poemsdataset

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